

Channel Capacity

Establish Relationship between :-

- System bandwidth
- Signal Bandwidth
- Signal impairment.

The maximum data rate at which information can be transmitted over a communication channel reliably without errors.

Two scientist theories:-

1. > Nyquist (No external noise, i.e. noise free channel)

Limitation on data rate is simply the bandwidth of the channel.

$$C = 2 B \log_2 M$$

Maximum Data Rate Measured in bits/second	Bandwidth of transmission system in Hz	Number of discrete states in digital signal.
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Now we run into a problem:-

As Data Rate increases

Bit error rate increases

becomes difficult for receiver to distinguish different signal state.

* Noise distorts a signal during transmission.

↳ greater Noise $\approx$ greater bit error rate for digital signal

* Signal To Noise Ratio (SNR):-

$$\underline{SNR} = \frac{S}{N}$$

average signal power

average noise power

SNR in Decibels (dB):-

$$SNR_{db} = 10 \log_{10} \left(\frac{S}{N} \right)$$

High SNR $\rightarrow$ Signal is stronger than Noise
↳ good Quality

decibels is just a logarithmic unit of ratios, it simplifies large ratios into easier formats

2. > Shannon's Noisy Channel

extended Nyquist's work and took into account the effects of noise.

$$\rightarrow C = B \log_2 (1 + S/N)$$

hence data rate is limited by bandwidth and Noise.

According to Shannon, Max data Rate achieved is determined by :-

- Bandwidth of the system/channel
- Noise on that channel.